

BUAN 6341.501 – Applied Machine Learning

Project Proposal

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Multi-Disease Classification from Chest X-Ray Images Using Deep Learning

Problem Description

Chest X-rays are one of the most frequently performed diagnostic imaging procedures worldwide, with over 2 billion examinations conducted annually. Radiologists must interpret these images to detect and diagnose various thoracic diseases, including pneumonia, cardiomegaly, pleural effusion, lung masses, and other pulmonary conditions. However, the interpretation process is time-intensive, requires specialized expertise, and is subject to human error – particularly in high-volume clinical settings or resource-limited regions where radiologist shortages exist. Missed diagnoses or delayed detection can lead to poor patient outcomes, while false positives create unnecessary anxiety and costly follow-up procedures.

This project addresses the critical need for automated, accurate, and efficient chest X-ray disease classification systems. By developing machine learning models capable of simultaneously detecting multiple thoracic pathologies from radiographic images, this work aims to support clinical decision-making through automated screening and diagnostic assistance. The proposed system would benefit radiologists by providing a “second opinion” to improve diagnostic accuracy, help prioritize critical cases requiring immediate attention, and enable faster turnaround times in emergency departments. Additionally, such systems could democratize access to expert-level diagnostic capabilities in underserved regions lacking specialized radiologists, ultimately improving patient care quality and health outcomes globally.

Data & Preprocessing

The dataset for this project consists of the *NIH ChestX-ray14* dataset, comprising 112,120 frontal-view chest X-ray images from 30,805 unique patients. Each high-resolution (1024x1024 pixel) grayscale image is labeled with one or more of 14 common thoracic disease classes: Atelectasis, Cardiomegaly, Effusion, Infiltration, Mass, Nodule, Pneumonia, Pneumothorax, Consolidation, Edema, Emphysema, Fibrosis, Pleural Thickening, and Hernia. The dataset also includes patient metadata such as age & gender. This multi-label classification problem reflects real-world clinical scenarios where patients often present with multiple concurrent conditions, making it both challenging & clinically relevant.

Preprocessing will involve multiple steps to prepare the data for model training. Images will be resized from 1024×1024 to resolution-appropriate sizes based on each architecture's specifications. Pixel intensity values will be normalized to the range, and histogram equalization techniques may be applied to enhance contrast and improve feature visibility. The dataset will be split into training (70%), validation (15%), and test (15%) sets, stratified by disease prevalence to ensure representative class distributions across splits.

Proposed Machine Learning Method

This project will implement state-of-the-art deep learning architecture, to thoroughly investigate the problem. The core methodology will leverage *Convolutional Neural Networks (CNNs)* fine-tuned for multi-label chest X-ray classification. Three different architectures will be implemented and compared: (1) **ResNet-101**, known for its residual connections that facilitate training of deep networks; (2) **DenseNet-121**, which has shown strong performance on chest X-ray tasks due to feature propagation; (3) **EfficientNet-B7**, which optimizes the trade-off between model complexity and performance. These models will be initialized with ImageNet pretrained weights, and the final fully connected layers will be modified to output 14 disease predictions using sigmoid activation functions for multi-label classification. Training will employ binary cross-entropy loss, the Adam optimizer with learning rate scheduling, and early stopping based on validation performance.

To potentially improve predictive performance beyond individual models, ensemble techniques will be explored. A *weighted average ensemble* combining predictions from *ResNet-101*, *DenseNet-121*, and *ElasticNet-B7* will be implemented, with weights optimized on the validation set. Additionally, if time permits, a *stacking ensemble* using a meta-learner trained on the outputs of base models will be investigated.

Evaluation Methodology

Given the multi-label nature of the problem and significant class imbalance, model evaluation will employ multiple complementary metrics. The primary evaluation metric will be *Area Under the Receiver Operating Characteristic Curve (AUC-ROC)* calculated independently for each of the 14 disease classes and then averaged (macro-average). *AUC-ROC* is particularly appropriate for imbalanced medical datasets as it measures discrimination ability across all decision thresholds. Additional metrics will include *precision*, *recall* and *F1-score* for each disease class at optimized thresholds, as well as overall accuracy and Hamming loss for multi-label prediction quality.

Model comparison will be conducted through statistical testing using *DeLong's test* for comparing AUC values between models. *Confusion matrices* and per-class performance breakdowns will provide insight into which diseases are most challenging to detect and where models excel or struggle. Visualization techniques including *Gradient-weighted Class Activation Mapping (Grad-CAM)* will be employed to interpret model predictions and understand which image regions influence disease classification decisions, providing clinical interpretability.

Cross-validation on the training set will be performed to tune hyperparameters including learning rate, batch size, dropout rates and data augmentation parameters. The final model comparison will be conducted on the held-out test set, ensuring unbiased performance estimates. Computational requirements will be managed through cloud resources (**Google Cloud Platform**) using **VSCode Remote-SSH** for development, enabling efficient experimentation with multiple architectures within reasonable time constraints.

Expected Outcomes & Impact

This project will produce a comprehensive comparison of machine learning approaches for multi-disease chest X-ray classification, with expected *AUC-ROC* performance in the range of *0.70 – 0.85 per disease class* based on published benchmarks. The analysis will provide insight into the relative strengths of different CNN architectures and the value of ensemble methods for complex medical imaging tasks. Beyond model performance metrics, this work will contribute understanding of the practical considerations for deploying automated diagnostic assistance systems, including computational requirements, interpretability needs, and performance trade-offs between model complexity and accuracy. Ultimately, the project demonstrates the potential for deep learning to augment clinical workflows, reduce diagnostic errors, and improve access to quality healthcare through intelligent automation of medical image interpretation.